

802.11be vs 802.11n: Rate-Matched P-EDCA Tail-Latency Comparison

CWds × QSRC × PSRC parameter sweep | CBR / Poisson / MMPP / OnOff traffic | P-EDCA STA perspective

Simulation setup

Topology	30 STAs → 1 AP, random 1-5 m disc; 5 GHz ch36 / 20 MHz
Traffic	Uplink AC_VO UDP, 1 Mbps per STA (1000 B); CBR / Poisson / MMPP / OnOff
PHY	11n = HtMcs7 @ 0.8 μs GI vs 11be = EhtMcs5 @ 1.6 μs GI — both 65.0 Mbps (rate-matched)
Airtime	1×1000 B PPDU: EHT 177 μs vs HT 160 μs (+10.6%); per-MPDU @16× A-MPDU +0.6%
Control	OFDM 6 Mbps for both; A-MPDU on, BA window 64 on both; VO TXOP limit 2.08 ms
Sweep	nPedca ∈ {5, 15, 30}; CWds{0,1} × QSRC{0-5} × PSRC{1-3} = 36 combos
Statistics	10 seeds × 10 s per point; plus an EDCA-only baseline (nPedca = 0)

Caveat: 11n data ran 2026-06-22 on v6.2.1 code; 11be re-ran 2026-07-18 on v6.3.2 (EHT NAV / frame-exchange fixes) — cross-standard deltas mix PHY and code-vintage effects.

P99 −46% ~ −22%

Full P-EDCA penetration (30/30), best params: 11be wins
11be 11.6–14.6 ms vs 11n 17–21 ms — only at n=30 (see p.2)

EDCA-only ×2.1–2.4

Without P-EDCA the rate-matched 11be tail is WORSE
P99 19–22 → 45–47 ms; carried VO throughput -18% ~ -12%

+69% ~ +75% vs EDCA

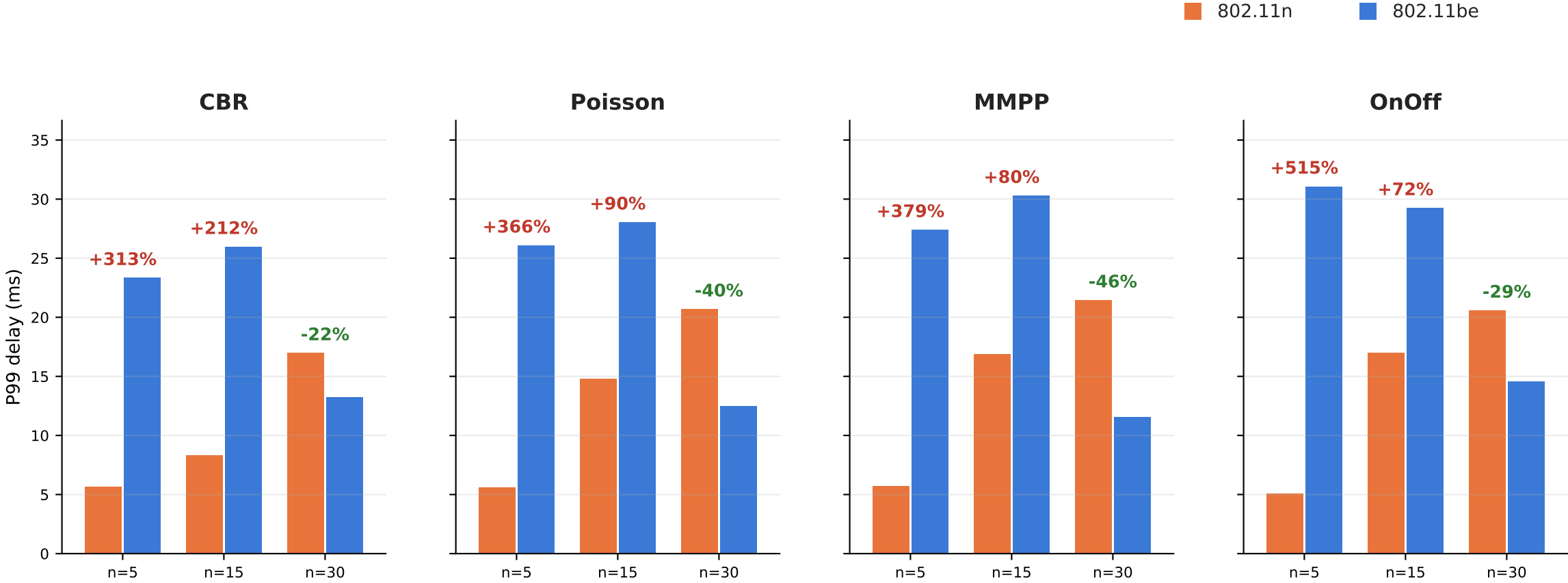
P-EDCA value at n=30 is huge on 11be, tiny on 11n
11n only +3% ~ +13% — the penetration trends run in OPPOSITE directions

One-line takeaway

At the SAME 65 Mbps PHY rate, 11be's fixed per-PPDU overhead makes its saturated EDCA baseline ~2× worse at P99; aggressive P-EDCA (QSRC0/PSRC3) reclaims that dead time and pushes fully-penetrated 11be 22–46% below the best 11n — but partial penetration still trails.

Core result: P-EDCA P99 latency — 11be only wins at FULL penetration

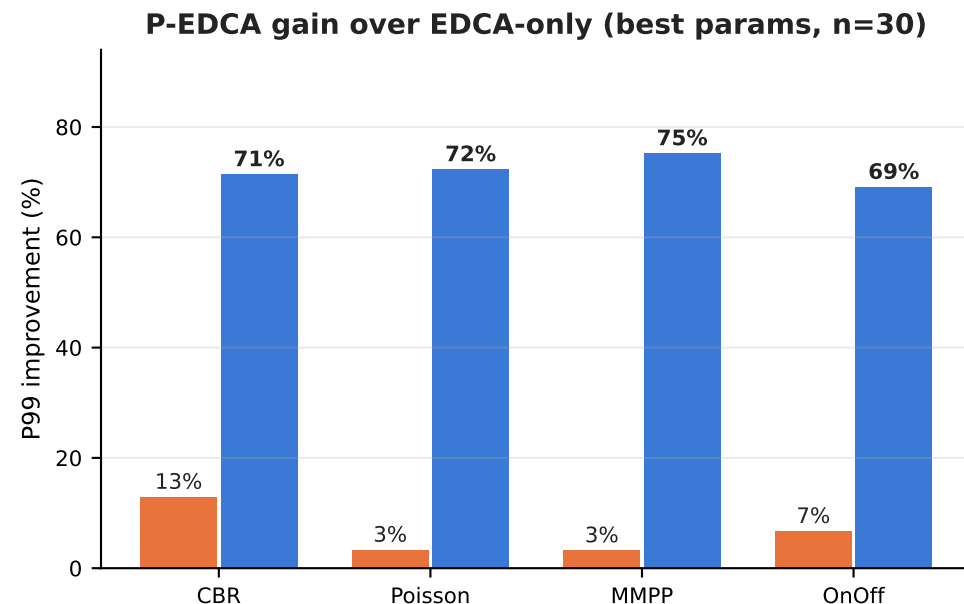
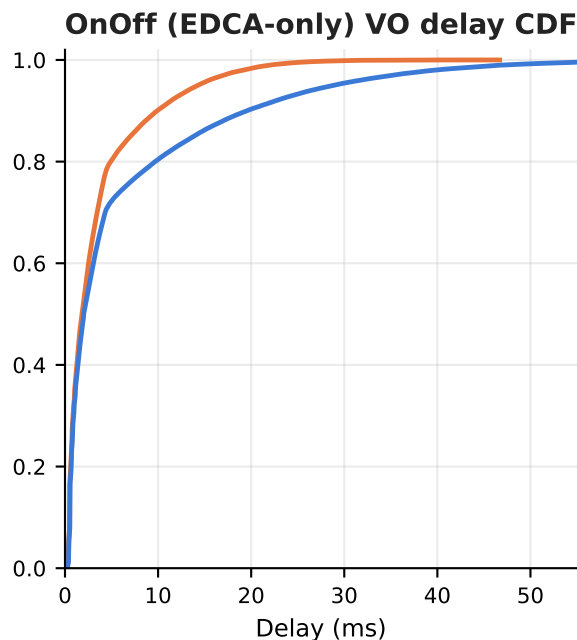
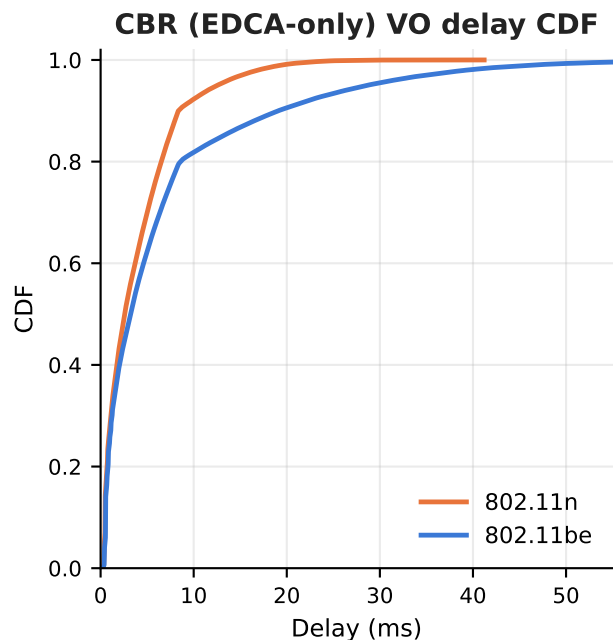
Minimum P99 over the 36-combo sweep, per traffic × nPedca (mean of 10 seeds; ms)



- Light penetration n=5: 11be is far WORSE (+313% to +515%). 11n's small P-EDCA subset rides a healthy baseline and reaches 5.1-5.7 ms; 11be's subset is dragged by its congested surroundings (23-31 ms).
- Medium n=15: still +72% to +212%. Full penetration n=30: the sign flips — 11be -22% to -46% (11n 17-21 ms vs 11be 11.6-14.6 ms).
- The n=30 win is parameter-dependent, not free: the MEDIAN over all 36 combos is still +72% to +123% (11be worse); only aggressive QSRC 0-1 + PSRC 3 flips the sign. All four 11be winners are c*q0s3.
- Gain-vs-own-EDCA runs in OPPOSITE directions with penetration: 11n +71~+77% at n=5 collapsing to +3~+13% at n=30 (priority dilutes); 11be +34~+50% rising to +69~+75%.

Mechanism: per-PPDU overhead amplified by saturation queueing

EDCA-only baseline (no P-EDCA, 30 STAs) isolates the two PHYs at the same 65 Mbps — 11be's whole distribution shifts right.

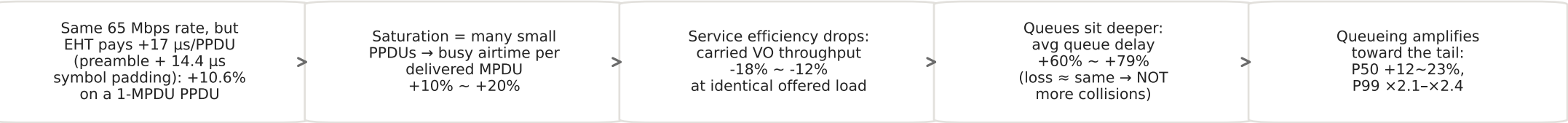


- The right-shift grows toward the tail: P10 +11~+11% ($\approx$ the longer EHT preamble), P50 +12~+23%, P99 +111~+137% — the signature of a queueing system pushed deeper into saturation, not of slower frames.
- The deficit widens with burstiness: carried VO throughput -12% (CBR) $\rightarrow$ -15% (Poisson) $\rightarrow$ -16% (MMPP) $\rightarrow$ -18% (OnOff); busy airtime per delivered MPDU +10% $\rightarrow$ +20% — burstier arrivals mean more small PPDU's, each paying the fixed EHT preamble tax.
- P-EDCA view: at n=30 the DS-CTS mechanism barely pays on 11n (+3~+13%) but recovers 11be's dead time (+69~+75%), landing 11be at 11.6–14.6 ms — below 11n's best.

Root cause: a fixed per-access tax amplified by queueing

Rate-matched at 65 Mbps — the difference is per-PPDU overhead and operating point, not per-bit speed

Causal chain (EDCA-only, 30 saturated VO STAs; measured values)



Supporting evidence (EDCA-only, 30 STAs; 11n → 11be)

Traffic	VO retry-limit loss	Queue delay (μs)	Access delay (μs)	Channel idle
CBR	9.1% → 8.8%	3636 → 6273	312 → 511	25.1% → 26.8%
Poisson	9.1% → 7.4%	3904 → 6237	366 → 544	23.4% → 25.4%
MMPP	8.9% → 7.7%	3904 → 6450	393 → 588	22.8% → 24.7%
OnOff	7.0% → 6.0%	3174 → 5672	421 → 660	22.4% → 23.8%

Reading: queue delay dominates MAC delay and rises 60–79%; access delay rises 49–64% (longer PPDU + response timing); idle even rises 1.4–2.0 pp — the channel is not busier, it is less efficient per access; retry-limit loss is slightly LOWER on 11be, ruling out a collision-rate explanation.

Open items (suggested follow-ups)

(1) Re-run the 11n sweep on current v6.3.2 code — the 11n data predates the v6.3.x P-EDCA/NAV fixes. (2) Decompose the per-PPDU tax vs EHT code-path effects: count PhyRxDrop / collisions per standard; re-run 11be at GI 0.8 μs (68.8 Mbps). (3) Log the A-MPDU size distribution to confirm the small-PPDU hypothesis behind the +10~20% busy-time per MPDU.

ns-3.45 P-EDCA sims | scratch/delay_pdf/{11n,11be} | 11n: 2026-06-22 (v6.2.1) · 11be: 2026-07-18 (v6.3.2)

Why each pattern appears

■ P-EDCA flips the ranking at n=30

DS-CTS ordering replaces CWmin=3 collision contention — exactly the dead time 11be loses; 11n's healthier baseline leaves little to reclaim.

■ Partial penetration favors 11n

The P-EDCA subset rides on the surrounding system: 11be's congested baseline drags its subset tail even with priority access.

■ Burstier traffic widens the gap

Bursts drain in small PPDU's; each PPDU pays the fixed EHT preamble tax, so the efficiency deficit grows (−12% CBR → −18% OnOff).